

ABDULLAH BAKR

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SUMMARY

AI/ML & Deep Learning Engineer with 3+ years of Python experience. Specializes in computer vision, transfer learning, and end-to-end machine learning pipelines. Built 7 real-world projects — including medical imaging AI, multi-class image classifiers, and an open-source data-cleaning library published on PyPI. Familiar with data visualization using Matplotlib and Seaborn. Exploring large-scale ML pipelines and generative AI techniques. Actively seeking internships and full-time AI/ML roles.

TECHNICAL SKILLS

Languages	Python (3+ years)
Deep Learning	TensorFlow, Keras, PyTorch — CNNs, Transfer Learning, custom training loops, regularization
Computer Vision	Image classification, data augmentation, OpenCV, ResNet50, VGG19, InceptionV3, EfficientNet, Swin Transformer
ML & Data Science	Scikit-learn, Pandas, NumPy, SciPy — EDA, regression, classification, feature engineering, cross-validation
Visualization	Matplotlib, Seaborn
Tools	Git, GitHub, Jupyter Notebook, VS Code, PyPI package publishing, Kaggle

PROJECTS

Breast Cancer Classification — Transfer Learning (Best: 98.5% Val Acc) 2026

github.com/abdubakr77/cancer-classification-TensorFlow

- Comparative study of ResNet50, VGG16, and EfficientNetB4 for breast cancer classification on ultrasound images.
- Best model (ResNet50) achieved 98.5% validation accuracy on 2-class prediction (benign / malignant).
- Stack: Python, TensorFlow, Keras, Jupyter Notebook.

Cats vs Dogs Image Classifier (~98% Test Accuracy) 2026

github.com/abdubakr77/Cats-vs-Dogs-Image-Classification-98-Acc

- Built a transfer learning model using ResNet50 pre-trained on ImageNet.
- Applied data augmentation (zoom, flip, shear, rotation, brightness) and EarlyStopping on validation loss.
- Stack: Python, TensorFlow, Keras, Jupyter Notebook.

Garbage Classification — InceptionV3 (~93% Accuracy) 2026

github.com/abdubakr77/garbage-classification-inception-v3

- Multi-class image classifier for waste sorting (cardboard, glass, metal, paper, plastic, trash) on ~25K images.
- Stack: Python, PyTorch, InceptionV3.

Intel Image Classification — Transfer Learning (~92.5% Val Acc) 2026

github.com/abdubakr77/intel-image-classification-cnn

- Multi-class scene classification (6 categories: Buildings, Forest, Glacier, Mountain, Sea, Street) on ~24K images.
- Used InceptionV3 with frozen pretrained base and custom classification head.
- Stack: Python, TensorFlow, Keras.

Breast Cancer Classification — VGG19-BN (92% Test Accuracy) 2026

github.com/abdubakr77/cancer-classification

- Deep learning model on ultrasound images (benign / malignant / normal) using VGG19 with batch normalization.

- Stack: Python, PyTorch, Jupyter Notebook.

DeepCSV — Automatic Data Cleaning Library (Published on PyPI)

2026

github.com/abdubakr77/deepcsv

- Built and published an open-source Python library on PyPI for automated CSV/XLSX data cleaning.
- Auto-walks directory trees, converts stringified lists to NumPy arrays, fixes mixed-type columns, exports to Parquet.
- Stack: Python, Pandas, NumPy, PyArrow. Installable via: `pip install deepcsv`.

Weather Temperature Prediction — Regression Pipeline

2026

github.com/abdubakr77/Weather-Temperature-Prediction

- End-to-end regression pipeline predicting temperatures from historical meteorological data.
- Feature engineering, cross-validation, model comparison: Linear Regression, Ridge, Random Forest.
- Stack: Python, Scikit-learn, Pandas, Matplotlib, Jupyter Notebook.

NIH Chest X-Ray Disease Classification — Research (In Progress)

2026

GitHub — Upcoming | Paper — Upcoming

- Multi-label classification of 14 thoracic diseases from 112K+ chest X-ray images (NIH ChestX-Ray14).
- Benchmarking DenseNet-121, ResNet-50, EfficientNet-B4, ViT-Base against a proposed Swin Transformer with Grad-CAM explainability.
- Targeting peer-reviewed publication. Stack: Python, PyTorch, Swin Transformer.

EDUCATION

High Institute for Engineering and Technology — Science Valley Academy

2025 – Present

B.Sc. in Management Information Systems (MIS) | Cairo, Egypt

CERTIFICATIONS

- AI for Everyone (AI4E) — ITI / Mahara-Tech, AI Academy
- Introduction to Modern AI — Cisco Networking Academy
- Python Programming Basics — ITI / Mahara-Tech, Cybersecurity Academy
- Boost Your Productivity with AI — Google / Maharat من Google
- Python & AI Course — Black Horse Courses, Eng. Omar Hany

LANGUAGES

Arabic (Native) | English (Intermediate)